A PROJECT TITLE ON

Accident Hotspot Prediction Using Random Forest and Spatial Clustering in Machine Learning

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING (25SC2107E)

BACHELOR OF TECHNOLOGY

IN

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DEPARTMENT OF CSE

Road traffic accidents remain a serious public safety concern, with certain locations experiencing a disproportionately high frequency of incidents. This project, "Accident Hotspot Prediction Using Random Forest and Spatial Clustering," aims to identify and predict accident-prone zones by combining machine learning with spatial data analysis. Historical accident records containing location, time, and environmental attributes are collected and preprocessed to ensure data quality and consistency. Spatial clustering techniques are applied to group geographically related accident points and reveal hotspot regions, while a Random Forest classifier is trained on relevant features to predict the likelihood of accidents in a given area. The performance of the model is evaluated using standard classification metrics such as accuracy, precision, and recall to assess its reliability. The resulting hotspot predictions can assist traffic authorities and urban planners in prioritizing safety interventions, optimizing patrol routes, and implementing preventive measures in high-risk areas. By integrating predictive analytics with spatial insights, this project demonstrates how data-driven approaches can contribute to safer road infrastructure and more effective accident prevention strategies.

2.Inroduction

This project uses historical accident data to identify accident-prone areas. The data is cleaned and processed before applying machine learning techniques. Spatial clustering is used to group nearby accident locations and identify areas with a high concentration of accidents. The Random Forest algorithm is then used to predict accident-prone locations. The results can help authorities understand accident patterns and take preventive measures.

3. SOFTWARE REQUIREMENTS:

- **Programming Language:** Python
- **Development Tool:** Jupyter Notebook / Google Colab / VS Code

- **Dataset:** Road Accident Dataset (CSV/Excel)
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Machine Learning:** Scikit-learn
- **Clustering:** K-Means / Spatial Clustering
- **Prediction Algorithm:** Random Forest Classifier
- **Visualization:** Matplotlib and Seaborn
- **Data Processing:** Pandas and NumPy

Signature of the Guide

Course Instructor